$$f = \frac{f_{CLK}}{PSC+1} = \frac{16\,MHz}{3+1} = 4\,MHz$$

$$T_{tick} = \frac{1}{f} = 0.25\,\mu s$$

$$ARR+1 = 7$$

$$T_{PMW} = (ARR+1)T_{tick}$$

$$= 7 \times 0.25$$

$$= 1.75\,\mu s$$

$$f = \frac{1}{T_{PMW}} = \frac{1}{1.75\,\mu s} = 571.4\,kHz$$

$$\text{Dute Cycle} = \frac{3}{6+1} = \frac{3}{7} \approx 42.86\%$$